Tutorial Sheet 3

Introduction to Numerical Methods for ODEs and PDEs

Systems of ODEs & Boundary Value Problems

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Exercise 1 (Pendulum)
The motion of a simple pendulum is described by the differential equation

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \frac{g}{l} sin(\theta) = 0\,,$$

where $g = 9.81$ is the gravitational constant, $l = 0.1$ is the length of the pendulum and θ is the angle of the pendulum with respect to the vertical axis. We wish to obtain a numerical solution for the interval $t \in [0, 5]$. Suppose the pendulum has an intial angle $\theta(0) = 1$ and an initial angular velocity $\theta'(0) = 0$.

$y_1 = \theta,\ y_2 = \theta' \quad \therefore y_1' = y_2,\ y_2' = -\frac{g}{l}\sin(y_1) \quad \therefore y_1^{(0)} = 1,\ y_2^{(0)} = 0 \quad \therefore y' = \begin{bmatrix} y_2 \\ -\frac{g}{l}\sin(y_1) \end{bmatrix},\ y(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

1. Write the above second order ODE as a system of (two) first order ODEs.
2. Use explicit Euler with stepsize $h = 0.01$ to obtain a numerical solution of the ODE. Comment on the resulting numerical solution.
3. Use 4th-order Runge-Kutta with stepsizes $h = 0.1$ and $h = 0.05$ to obtain numerical solutions of the ODE. Comment on the resulting numerical solutions.
4. Obtain and plot a numerical solution using MATLAB's built-in function "ode45" with the default tolerance options.

Exercise 2 (Simple harmonic oscilaltor)
Consider the simple harmonic oscillator

$$y' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} y = Ay\,. \quad = \begin{bmatrix} y_2 \\ -y_1 \end{bmatrix}$$

In Tutorial Sheet 1 we saw that the system of ODEs can be converted, via a change of coordiantes, to two uncoupled scalar ODEs. In this exercise we wish to solve the system of ODEs directly (without performing any change of coordinates). Let $y(0) = \begin{bmatrix} 10 \\ 1 \end{bmatrix}$.

1. Attempt to obtain a numerical solution using explicit Euler with stepsize $h = 0.1$. What do you observe and why?
2. Use 4th-order Runge-Kutta with stepsize $h = 0.1$ to obtain a numerical solution of the system of ODEs.
3. Use the trapezoidal method (together with the inbuilt MATLAB function "fsolve") with stepsize $h = 0.1$ to obtain a numerical solution of the system of ODEs.
4. How do 4th-order Runge-Kutta and the trapezoidal method compare?

Exercise 3 (The van der Pol equation - a stiff system)
The ODE

$$x'' - \mu(1 - x^2)x' + x = 0\,,$$

where $\mu > 0$ is a scalar parameter, is the so-called van der Pol equation. Suppose $x(0) = 2$ and $x'(0) = 0$. We wish to obtain a numerical solution for the time interval $t \in [0, 50]$.

$y_1 = x,\ y_2 = x' \quad \therefore y_1' = y_2,\ y_2'' = -y_1 + \mu(1-y_1^2)y_2$

1. Write the second order ODE as a system of first order ODEs. $\therefore y' = \begin{bmatrix} y_2 \\ -y_1 + \mu(1-y_1^2)y_2 \end{bmatrix}$
2. Let $\mu = 1$ and obtain a numerical solution to the ODE using explicit Euler with stepsize $h = 0.01$. $y(0) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$
3. Again, letting $\mu = 1$, obtain a numerical solution using MATLAB's builti-in function "ode45".
4. Let $\mu = 2000$ and obtain a numerical solution using explicit Euler with the same stepsize used in part 2. of this exercise. What do you observe?
5. For $\mu = 2000$, we wish to obtain numerical solutions using MATLAB's built-in functions[1].
   (a) Obtain a numerical solution using "ode45" (with the default tolerance options).
   (b) Obtain a numerical solution using "ode15s" (with the default tolerance options).
   (c) Comment on the differences between the two.

Exercise 4 (A two-point boundary value problem)
Consider a thin metal rod of length $L = 4$ in a room with an ambient temperature $\bar{T} = 20$. At one end ($x = 0$), the rod is in contact with a material of constant temperature $T_0 = 10$. At the other end ($x = L$) the rod is in contact with a material of constant temperaturte $T_L = 36$. The temperature distribution $T(x)$ along the rod is described by the ODE

$$\frac{\mathrm{d}^2 T}{\mathrm{d}x^2} + \alpha(\bar{T} - T) = 0\,,$$

with boundary conditions $T(0) = T_0$ and $T(L) = T_L$.

Obtain a numerical solution for this boundary value problem using the shooting method (along with your preferred numerical scheme introduced for initial value problems).

$y_1 = T,\ y_2 = \frac{dT}{dx}$

$\therefore y_1' = y_2,\ y_2' = \alpha(y_1 - \bar{T}) = \alpha(y_1 - 20) \qquad \therefore y' = \begin{bmatrix} y_2 \\ \alpha(y_1 - 20) \end{bmatrix}$

[1] While you are asked to use the default options for the built-in functions, for your own interest, you may wish to experiment with different absolute/relative tolerances.